OSHA Flammable Liquid Storage Checklist (29 CFR 1910.106)

For on-site checks of flammable liquid containers, storage tanks and piping.

OSHA 29 CFR 1910.106

DATE	SHIFT
INSPECTOR	AREA / EQUIPMENT ID

Before you start

1. Mark every item OK, Defect or N/A.
2. For any Defect, write down what you found, tell the person responsible for the area, and check the item again once it is fixed.
3. The manufacturer's instructions and site rules may add checks. This list covers paragraphs (a) to (d)(1) only, not the later rules in the full text.

Legal basis: 29 CFR 1910.106: Flammable liquids (Part 1910 Subpart H, Hazardous Materials) Source: eCFR, Title 29, text as of 2026-11-09
Online version and sources: https://safetyindex.blog/us/standards/1910-106-flammable-liquids

SECTION A | Containers and ignition sources

No.	Check item	What to look for	OK	Defect	N/A	Notes
1	Storage vessels Ref. 1910.106(a)(32)	Flammable liquids are held only in a tank or in a container that meets the (d)(2) container rules.	☐	☐	☐	
2	Safety cans Ref. 1910.106(a)(29)	Cans used as safety cans are approved, hold 5 gallons or less, and have a working spring-closing lid and spout cover.	☐	☐	☐	
3	Ignition sources Ref. 1910.106(b)(6)	Where vapors may be present, open flames, smoking, cutting and welding, hot surfaces and spark sources are removed or controlled.	☐	☐	☐	
4	Ventilation Ref. 1910.106(a)(31)	Where ventilation is relied on, it keeps vapor-air mixtures from building up above one-fourth of the lower flammable limit.	☐	☐	☐	

SECTION B | Outside aboveground tanks

No.	Check item	What to look for	OK	Defect	N/A	Notes
5	Shell spacing Ref. 1910.106(b)(2)(ii)(a)	No two flammable liquid tanks are closer than 3 feet shell to shell.	☐	☐	☐	
6	LPG separation Ref. 1910.106(b)(2)(ii)(f)	LPG containers are 20 feet or more from flammable liquid tanks, unless an exception in the paragraph applies.	☐	☐	☐	
7	Dike walls Ref. 1910.106(b)(2)(vii)(c)(3)	Walls are earth, steel, concrete or solid masonry, intact and liquidtight, with no breaches or open drains.	☐	☐	☐	
8	Inside the dike Ref. 1910.106(b)(2)(vii)(c)(6)	No loose combustible material and no drums or barrels, empty or full, inside the diked area.	☐	☐	☐	
9	Emergency relief Ref. 1910.106(b)(2)(v)(a)	Each tank has a construction or device that relieves internal pressure from an exposure fire.	☐	☐	☐	

General guidance only, not legal advice. Check the current regulation and your employer's procedures.

SECTION B | Outside aboveground tanks

No.	Check item	What to look for	OK	Defect	N/A	Notes
10	Vent device marking Ref. 1910.106(b)(2)(v)(h)	Each commercial venting device is stamped with its opening pressure, full-open pressure and flow capacity.	☐	☐	☐	
11	Vent outlet location Ref. 1910.106(b)(2)(vi)(b)	For Category 1, 2 or lower-flashpoint Category 3 tanks next to buildings or public ways, outlets are 12 feet or higher above ground, 5 feet or more from building openings, and not under eaves.	☐	☐	☐	
12	Gaging openings Ref. 1910.106(b)(2)(viii)(d)	Every gaging opening has a vaportight cap or cover in place.	☐	☐	☐	
13	Fill and emptying connections Ref. 1910.106(b)(2)(viii)(f)	Connections are outside buildings, clear of ignition sources, 5 feet or more from building openings, closed and liquidtight when idle, and identified.	☐	☐	☐	
14	Steel supports Ref. 1910.106(b)(5)(ii)	Steel supports have fire-resistive protection rated at 2 hours or more, or water spray, unless they are saddles under 12 inches high.	☐	☐	☐	

SECTION C | Underground and indoor tanks

No.	Check item	What to look for	OK	Defect	N/A	Notes
15	Underground vents (lower flashpoints) Ref. 1910.106(b)(3)(iv)(a)	Vent outlets are outside buildings, higher than the fill pipe opening, 12 feet or more above ground, and discharge upward only.	☐	☐	☐	
16	Underground vents (higher flashpoints) Ref. 1910.106(b)(3)(iv)(c)	Vents for higher-flashpoint Category 3 or Category 4 tanks end outside the building, above the fill pipe opening and above normal snow level.	☐	☐	☐	
17	Underground gaging openings Ref. 1910.106(b)(3)(v)(b)	Openings separate from the fill pipe have a liquid-tight cap. If inside a building, they also have a spring loaded check valve or other approved device.	☐	☐	☐	
18	Indoor tank vents Ref. 1910.106(b)(4)(ii)	Vents from indoor tanks discharge vapors outside the building.	☐	☐	☐	
19	Heat-actuated valves Ref. 1910.106(b)(4)(iv)(c)	Each withdrawal connection below the liquid level has an automatic-closing heat-actuated valve, unless it is an emergency disposal connection or the building is one-story and designed and protected for flammable liquid storage.	☐	☐	☐	
20	Indoor fill inlet Ref. 1910.106(b)(4)(iv)(g)	The fill inlet is outdoors, clear of ignition sources, 5 feet or more from building openings, capped liquidtight and identified.	☐	☐	☐	
21	Overflow prevention Ref. 1910.106(b)(4)(iv)(h)	Indoor tanks have a device or other means that stops overflow into the building.	☐	☐	☐	

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SECTION D | Piping and valves

No.	Check item	What to look for	OK	Defect	N/A	Notes
22	Pipe joints Ref. 1910.106(c)(3)	Joints are welded, screwed or approved connectors, with no drips or wet spots.	☐	☐	☐	
23	Friction joints indoors Ref. 1910.106(c)(3)	No joints inside buildings that rely on the friction of combustible materials to hold together.	☐	☐	☐	
24	Supports and protection Ref. 1910.106(c)(4)	Piping is firmly supported and protected from physical damage, settlement, vibration, expansion and contraction.	☐	☐	☐	
25	Corrosion protection Ref. 1910.106(c)(5)	Piping exposed to external corrosion is painted or otherwise protected, with no bare rusting sections.	☐	☐	☐	
26	Backflow check valve Ref. 1910.106(c)(6)	Where tank cars or vehicles pump into storage and backflow is possible, the connection has a check valve.	☐	☐	☐	
27	Pressure test Ref. 1910.106(c)(7)	New or altered piping was tested before use: hydrostatic to 150 percent or pneumatic to 110 percent of maximum anticipated pressure, at least 5 psig at the high point, held 10 minutes or longer.	☐	☐	☐	

DEFECTS FOUND AND ACTION TAKEN

Item no.	Defect description	Reported to	Action taken / date closed

SIGN-OFF

SAFE TO OPERATE? ☐ Yes ☐ No: tagged out of service	
INSPECTOR SIGNATURE	DATE / TIME
SUPERVISOR REVIEW	DATE / TIME